

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: S Y B Tech Branch : Chemical Engineering Semester : III

Subject Code & Name: BTCHC303 Process Calculations

Max Marks: 60

Date: 10/02/2025

Duration: 3 Hr.

Instructions to the Students:

- Each question carries 12 marks.
- Question No. 1 will be compulsory and include objective-type questions.
- Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
- The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- Use of non-programmable scientific calculators is allowed.
- Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	44.8 Nm ³ /h of air is equivalent to kmol/h of air.	2/CO2	1
	a. 1 b. 1.5 c. 2 d. 2.5		
2	. The mass fraction of methanol in an aqueous solution is 0.64. The mole fraction of methanol (x_{Me}) satisfies	4/C3	1
	a. $x_{Me} < 0.5$ b. $x_{Me} = 0.5$ c. $0.5 < x_{Me} < 0.64$ d. $x_{Me} > 0.64$		
3	Assuming that CO ₂ obeys perfect (ideal) gas law, calculate the density (ρ) of CO ₂ (in kg/m ³) at 263° C (536.15 K) and 2 atm	4/CO2	1
	a. 1 b. 2 c. 3 d. 4		
4	Quantities of water to be evaporated in concentrating NaOH solution from 10% to 20% and from 20% to 30% are	1/CO2	1
	a. same b. different c. almost same d. none of these		
5	A material which does not change in terms of its quantity during the particular operation, is called material for material balance.	2/CO3	1
	a. constant b. main c. tie d. unreacted		
6	Pure A in gas phase enters a reactor. 50% of this A converted to B through the reaction $A = 3B$. Mole fraction of A in the exit stream is	3/CO5	1
	a. 1/2 b. 1/3 c. 1/4 d. 1/5		
7	An equimolar liquid mixture of species 1 and 2 is in equilibrium with its	3/CO3	1

	vapour at 400 K. At this temperature, the vapour pressures of the species are $p_1^{\text{sat}} = 180 \text{ kPa}$ and $p_2^{\text{sat}} = 120 \text{ kPa}$. Assuming that Raoult's law is valid, the value of y_1 is		
	a. 0.30 b. 0.41 c. 0.60 d. 0.50		
8	For an equilibrium binary vapour-liquid mixture, vapour is at temperature and liquid is at temperature.	3/CO4	1
	a. condensation, boiling b. dew point, bubble point c. condensation, bubble point d. dew point, boiling		
9	Heat capacity while latent heat with increase in temperature	4/CO4	1
	a. increases, decreases b. increases, increases c. decreases, decreases d. decreases, increases		
10	If carbon monoxide appears in flue gas, combustion is termed as combustion.	4/CO4	1
	a. complete b. about to complete c. partial d. None of these		
11	Adiabatic saturation temperature is wet bulb temperature.	5/CO4	1
	a. equal to b. greater than c. less than or equal to d. greater than or equal to		
12	When a solid or gas is dissolved in the solvent, the heat evolved or absorbed is called	2/CO4	1
	a. heat of solution b. heat of mixing c. excess enthalpy d. Both (a) & (c)		
Q. 2	Solve Any Two the following.		12
A)	Define Normality, Molarity and Molality of a solution	2/CO1	6
B)	The volumetric flow rate of CCl_4 in a 1.0-cm-diameter pipe is $100 \text{ cm}^3/\text{min}$ (molecular weight of $\text{CCl}_4 = 153.838$ and density of $\rho_{\text{CCl}_4} = 1.595 \text{ g/cm}^3$.) (i). What is the mass flow rate? (ii) What is the molar flow rate? (iii) What is the linear velocity of CCl_4 ?	2/CO2	6
C)	A gas mixture contains 0.28 kmol of HCl , 0.34 kmol of N_2 and 0.09 kmol of O_2 . Calculate: (i) Average molecular weight of gas (ii) Volume	2/CO2	

	occupied by this mixture at 4 atm and 303 K.		
Q.3	Solve Any two the following.		12
A)	Soyabean seeds oil is extracted with hexane in a batch extractors. The flaked seeds contain 18.2% oil, 69.5% solid and 12.3% moisture. At the end of the process, cake is separated from hexane oil mixture. The cake analysis yields 0.8% oil, 88.2% solids and 11.0% moisture. Find the percentage recovery of oil. All percentage are by weight.	2/CO2	6
B)	C_2H_4O is manufactured by the oxidation of ethylene. 120 kgmol of C_2H_4 is fed to the reactor. The product found to contain 90 kgmol of ethylene oxide and 8 kgmol of CO_2 . Calculate: (i) Percentage conversion of ethylene (ii) Percentage yield of C_2H_4O .	1/CO2	6
C)	i) Write the material balance equation for an evaporator explaining the terms. ii) Write the material balance equation for a crystallizer explaining the terms iii) Write the material balance equation for a distillation column explaining the terms.	1/CO3	
Q. 4	Solve Any Two of the following.		12
A)	An air (B)–water (A) sample has a dry bulb temperature $55^\circ C$ and an absolute humidity 0.030 kg water/kg dry air at 1 std. atm pressure. Calculate: (i) The percentage humidity (ii) The molal absolute humidity (iii) The partial pressure of water vapour in the sample Given Saturated humidity = 0.115	2/CO3	6
B)	Calculate the total pressure and the composition of vapour in contact with a solution at $100^\circ C$, containing 35% Benzene, 40% Toluene and 25% Ortho-xylene by weight. Data: Vapour pressure at $100^\circ C$ Benzene = 1340 mmHg ,Toluene = 560 mmHg ,O-Xylene = 210 mmHg	2/CO4	6
C)	Define i) Humidity ii) Percentage Humidity iii) Relative Humidity iv) Dew point v) Dry bulb temperature vi) wet bulb temperature	2/CO4	6

Q.5	Solve Any Two of the following.		12										
A)	A steam of carbon dioxide flowing at rate of 100 mol/min is heated from 298 K to 383 K. Calculate the heat that must be transferred using C_p°. Data: $C_p^\circ = a + b T + c T^2 + d T^3$, kJ/ (kmol.K) <table><tr><td>Gas</td><td>a</td><td>$b \times 10^3$</td><td>$c \times 10^6$</td><td>$d \times 10^9$</td></tr><tr><td>CO₂</td><td>21.3655</td><td>64.2341</td><td>-41.0506</td><td>9.7999</td></tr></table>	Gas	a	$b \times 10^3$	$c \times 10^6$	$d \times 10^9$	CO ₂	21.3655	64.2341	-41.0506	9.7999	1/CO4	6
Gas	a	$b \times 10^3$	$c \times 10^6$	$d \times 10^9$									
CO ₂	21.3655	64.2341	-41.0506	9.7999									
B)	Calculate the standard heat of formation of n-propanol (C₃H₇OH) liquid using the following data: Standard heat of formation of CO₂ (g) = -393.51 kJ/mol Standard heat of formation of H₂O(l) = -285.83 kJ/mol Standard heat of combustion of n-propanol liquid = -2028.19 kJ/mol	4/CO4	6										
C)	Calculate the standard heat of reaction of the following reaction $C_5H_{12}(l) + 8O_2(g) \rightarrow 5CO_2(g) + 6H_2O(l)$ Data: Component ΔH_f(kJ/mol) C₅H₁₂ (l) = -173.49 , CO₂ (g) = -393.51 H₂O(l) = -285.83 , O₂ (g) = 0	2/CO4	6										
Q. 6	Solve Any Two of the following.		12										
A)	A tank containing 100 kg of a 60% brine (60% salt) is filled with a 10% salt solution at the rate of 10 kg/min. Solution is removed from the tank at the rate of 15 kg/min. Assuming complete mixing, find the kilograms of salt in the tank after 10 min.	2/CO5	6										
B)	5 m³ of a 20 mass% solution of sodium chloride contain in a tank. Water is added at the rate of 3 m³ /h and the solution is withdrawn at the same volumetric rate. Determine the mass of salt remaining after one hour. Assume through mixing in the tank and neglect density changes due to concentration. The tank can hold only 5 m³ of solution. How long will it take for the mass per cent of the solute to become 5?	1/CO5	6										
C)	Distinguish between a steady-state and an unsteady-state process.	4/CO5	6										
	*** End ***												